

Thermal arithmetic orbit weights and the shifted-zeta equation

Bost–Connes coefficients and the causal realization problem

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Abstract

The arithmetic orbit coefficients in the shifted completed-zeta quotient admit an exact thermal description. In the Bost–Connes system, a fixed primitive projection P_n has equilibrium expectation $\varphi_\beta(P_n) = \prod_{\ell|n} (1 - \ell^{-\beta})$. With inverse temperature $\beta = 2\omega$, the required coefficient is $c_n(\omega) = n^{(\beta-1)/2} \varphi_\beta(P_n)$. This law reproduces primes, repetitions, composites and the prime part of the shift generator. We derive the identity, an explicit positive finite-place model, and the local ordinary Hilbert-space norms. The centering factor and logarithmic delay interpretation still require a physical input/output construction. Two immediate assemblies fail: the bare Euler delay train is noncontractive, and the natural thermal half-density vectors do not converge in the Haar-based infinite tensor product. These failures do not exclude the established equilibrium states or a different coupled realization. We formulate a two-prime coupling test and close with a brief account of earlier unsuccessful readouts.

Status. Exact coefficient matching; causal completed physical realization open. Prepared with LLM assistance; the existing 87-case diagnostic and review scope are documented in Appendix A.

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1 Thermal orbit weights: the current result

The current investigation seeks an independently defined physical system whose response supplies the arithmetic side of the shifted-zeta differential equation. Its most useful new result concerns the changing orbit weights. The Bost–Connes system has native projections that detect primitiveness at specified primes. Their positive thermal expectations, with one common inverse temperature, give precisely the product entering every required arithmetic coefficient. No separate choice is made for each prime or each repeated orbit.

More precisely, let ℓ always denote a prime, let φ_β be a Kubo–Martin–Schwinger (KMS) equilibrium state at inverse temperature β , and let P_n select states primitive at every prime dividing n . The dictionary is

$$\boxed{\beta = 2\omega, \quad c_n(\omega) = n^{(\beta-1)/2} \varphi_\beta(P_n), \quad \varphi_\beta(P_n) = \prod_{\ell|n} (1 - \ell^{-\beta}).} \quad (1.1)$$

The primitive projector is independent of temperature. The displayed centering factor is additional and explicit; its physical origin is not claimed to follow merely from the existence of a thermal state.

The established Bost–Connes theory supplies positive KMS states in the range $0 < \beta \leq 1$, although the usual global Gibbs trace does not converge there [1, 2]. This is the range needed to move from the modular half shift $\omega = 1/2$ toward zero. Thus the candidate is more informative than a formal identification of a zeta partition function. A finite primitive observable has a meaningful positive expectation at the relevant temperatures.

The contribution of this investigation is the exact comparison with the desired orbit coefficients and the analysis of specific readout and norm requirements. The underlying equilibrium system, KMS existence theorem and primitive-count identities are established mathematics. There is no claim of literature novelty for them.

Three levels must remain separate. First, the primitive probabilities in (1.1) are native thermal observables. Second, their centered values reproduce the Euler coefficients, including the entire prime part of the shift generator. Third, a causal physical response with these amplitudes, the archimedean completion and the ordinary radiation norm has not been constructed. Positivity at the first level does not prove the third.

Sections 2–3 establish the arithmetic dictionary. Section 4 gives the local Hilbert-space realization and two scoped assembly exclusions. The next finite-place physical test is stated in Section 5. Earlier WZW, Brownian and cusp constructions are summarized only near the end, in Section 6, with references to the detailed research notes. The current calculation is recorded in the orbit-weight note [4]; the present manuscript develops that investigation rather than treating the earlier tests as its organizing framework.

2 The arithmetic response and its orbit coefficients

2.1 The shift equation and the physical requirement

Define the completed function and its centered logarithmic derivative by

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad H(p) = \xi\left(\frac{1}{2} + p\right), \quad m(p) = \frac{H'(p)}{H(p)}. \quad (2.1)$$

The target transfer satisfies

$$K_\omega(p) = \frac{H(p-\omega)}{H(p+\omega)}, \quad \partial_\omega K_\omega = -a_\omega K_\omega, \quad a_\omega(p) = m(p-\omega) + m(p+\omega), \quad K_0 = 1. \quad (2.2)$$

These identities are initially used on a sufficiently far right, zero-free half-plane, and then meromorphically. Here p is the Laplace variable, ω the arithmetic shift, and β the thermal parameter in (1.1); physical time is a separate variable.

A physical realization must specify its states, preparation, propagation and readout, and derive K_ω from them. In particular, on an ordinary radiation space $L^2(0, L)$ its causal compression $V_{\omega, L}$ must have the appropriate identity limit and accumulated norm balance. The desired contraction is

$$\|V_{\omega, L} f\|_2^2 \leq \|f\|_2^2, \quad V_{0, L} = I. \quad (2.3)$$

The strong identity limit is the relevant convention; an operator-norm derivative at zero is not presumed. Cumulative contraction does not require every shift increment to be contractive, or a_ω itself to be a holomorphic positive-real function. Those distinctions and the unconditional weighted bounds are developed in [7].

Writing down the quotient in (2.2) is not a proof of the full unweighted realization. Its unit boundary modulus, where defined, follows from the functional equation and conjugation; causal contractivity for every positive shift is a further issue, related to the RH-equivalent innerness criterion [3]. No such conclusion is inferred here from the thermal coefficient identity.

2.2 Euler coefficients at unchanged logarithmic labels

For $\text{Re } p > 1/2 + \omega$, factor the target as

$$K_\omega(p) = A_\omega(p) Z_\omega(p), \quad Z_\omega(p) = \frac{\zeta(p + 1/2 - \omega)}{\zeta(p + 1/2 + \omega)}, \quad (2.4)$$

$$A_\omega(p) = \pi^\omega \frac{\Gamma(s_-/2)}{\Gamma(s_+/2)} \frac{s_-(s_- - 1)}{s_+(s_+ - 1)}, \quad s_\pm = p + \frac{1}{2} \pm \omega. \quad (2.5)$$

For one prime, set $a = \ell^{\omega-1/2}$, $b = \ell^{-\omega-1/2}$ and $z = \ell^{-p}$. Then

$$\frac{1 - bz}{1 - az} = 1 + \sum_{r \geq 1} (1 - \ell^{-2\omega}) \ell^{r(\omega-1/2)} z^r. \quad (2.6)$$

Multiplying the absolutely convergent local series gives

$$Z_\omega(p) = \sum_{n \geq 1} c_n(\omega) e^{-p \log n}, \quad c_n(\omega) = n^{\omega-1/2} \prod_{\ell|n} (1 - \ell^{-2\omega}), \quad c_1 = 1. \quad (2.7)$$

At $\omega = 1/2$, this is $c_n = \varphi_E(n)/n$, with φ_E Euler's totient. The orbit terminology refers to the primitive residues of the modular constant term, not to the closed-geodesic coefficients of a Selberg trace formula. The labels $\log n$ stay fixed as the arithmetic weights vary. Whether they are physical delays in a new model must be derived separately.

The first-prime coefficient and its tangent are

$$c_2(\omega) = \frac{2^\omega - 2^{-\omega}}{\sqrt{2}}, \quad c_2(1/2) = \frac{1}{2}, \quad c'_2(1/2) = \frac{3}{2} \log 2. \quad (2.8)$$

The full logarithmic derivative also follows in the same convergence region:

$$\partial_\omega \log Z_\omega(p) = 2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \cosh(\omega \log n) e^{-p \log n} = -a_{\omega, \text{prime}}(p). \quad (2.9)$$

The transfer involves all positive integers through multiplicativity; its logarithmic source involves prime powers through the von Mangoldt function Λ . The remaining terms of $-a_\omega$ are $\partial_\omega \log A_\omega$. A successful physical deformation must produce those terms with the same parameter.

3 Primitive projections in the Bost–Connes system

3.1 The equilibrium identity

The Bost–Connes arithmetic system was introduced in [1]. We use its multiplicative isometries v_n and the range projections $e_n = v_n v_n^*$, with

$$v_n^* v_n = I, \quad v_m v_n = v_{mn}, \quad e_m e_n = e_{\text{lcm}(m,n)}, \quad \sigma_t(v_n) = n^{it} v_n. \quad (3.1)$$

In the standard integer representation $v_n|k\rangle = |nk\rangle$, the projection e_n detects divisibility by n , and the Hamiltonian is $\mathcal{H}|k\rangle = (\log k)|k\rangle$. This representation has a normalized Gibbs trace only for $\beta > 1$. The full system nevertheless has positive KMS states for $0 < \beta \leq 1$; their existence and local measure structure are established inputs, for which we use [2, Introduction and Section 2].

For analytic elements the KMS condition reads $\varphi_\beta(ab) = \varphi_\beta(b\sigma_{i\beta}(a))$. Apply it to $a = v_n$ and $b = v_n^*$:

$$\varphi_\beta(e_n) = \varphi_\beta(v_n^* \sigma_{i\beta}(v_n)) = n^{-\beta}. \quad (3.2)$$

No analytic continuation of a divergent trace is used in this identity.

Proposition 3.1 (Primitive thermal coefficient identity). *For $n \geq 1$ define the temperature-independent projection*

$$P_n = \prod_{\ell|n} (I - e_\ell), \quad P_1 = I. \quad (3.3)$$

For a KMS state as above,

$$\varphi_\beta(P_n) = \prod_{\ell|n} (1 - \ell^{-\beta}). \quad (3.4)$$

Consequently the centered expectations at $\beta = 2\omega$ are exactly the coefficients in (2.7).

Proof. The divisibility projections commute. Expanding the finite product in (3.3) and using the lcm relation gives

$$P_n = \sum_{d|\text{rad}(n)} \mu(d) e_d, \quad (3.5)$$

where $\text{rad}(n)$ is the product of the distinct prime divisors, μ is the Möbius function and $e_1 = I$. Taking expectations and applying (3.2) gives the product in (3.4). Multiplication by $n^{(\beta-1)/2}$ gives (2.7). Thus the result does not require a separate probabilistic independence assumption. $\square$

At a repeated prime, $P_{\ell^r} = P_\ell$: primitiveness is the same event. Its contribution is

$$c_{\ell^r}(\omega) = (1 - \ell^{-2\omega}) \ell^{r(\omega-1/2)}. \quad (3.6)$$

The repetition dependence comes from centering; one has not introduced a new primitive event for each r . For coprime integers the projections combine by their distinct prime sets, recovering the required multiplicativity. Hence the first-prime tangent (2.8) and the full prime source (2.9) follow from the same coefficient law.

There is also a direct counting interpretation at integer exponent. Inclusion-exclusion shows that

$$J_d(n) = n^d \prod_{\ell|n} (1 - \ell^{-d}) \quad (3.7)$$

counts tuples $(x_1, \dots, x_d)$ modulo n with $\gcd(n, x_1, \dots, x_d) = 1$. For real $\beta > 0$ the analogous product defines $J_\beta(n)$, and $c_n(\omega) = J_{2\omega}(n)/n^{\omega+1/2}$. The thermal construction supplies a real-parameter probability law; it does not posit a noninteger number of discrete coordinates.

3.2 An explicit positive model at finitely many places

Let S be a finite set of primes. On $\bigotimes_{\ell \in S} \ell^2(\mathbb{N}_0)$, with occupation operators N_ℓ , take the self-adjoint Hamiltonian

$$\mathcal{H}_S = \sum_{\ell \in S} (\log \ell) N_\ell. \quad (3.8)$$

For every $\beta > 0$ its Gibbs state is trace class, with

$$\mathcal{Z}_S(\beta) = \prod_{\ell \in S} (1 - \ell^{-\beta})^{-1}, \quad \Pr_\beta(N_\ell = m) = (1 - \ell^{-\beta}) \ell^{-m\beta}. \quad (3.9)$$

The primitive projection selects occupation zero at each prime dividing n , provided those primes belong to S . It gives (3.4) directly. This finite-place realization of the relevant diagonal observables is explicit and positive throughout the temperature range of interest. It is not asserted to encode every observable of the full arithmetic algebra.

As a small illustration, the centered coefficients are

n	$\omega = 1/4$	$\omega = 0.4$	$\omega = 1/2$
2	0.246292858	0.397146260	1/2
3	0.321144348	0.523917402	2/3
4	0.207106781	0.370550563	1/2
6	0.079095559	0.208071837	1/3

The entries are samples of the exact identity, not a fit defining the state. In the infinite system for $\beta \leq 1$, the product partition function diverges. The positive KMS state must then be used in its established algebraic sense, not replaced by $e^{-\beta\mathcal{H}}/\zeta(\beta)$.

3.3 What the centering and the time variable still require

For $0 < \beta \leq 1$, the centered observable

$$B_{n,\beta} = n^{(\beta-1)/2} P_n \quad (3.10)$$

is a positive contraction and $\varphi_\beta(B_{n,\beta}) = c_n(\beta/2)$. However, this observable depends on temperature. The equality alone does not derive all amplitudes from a temperature change with a fixed physical detector.

The dynamics gives a structured analytic-time expression:

$$c_n(\beta/2) = n^{-1/2} \varphi_\beta \left(P_n v_n^* \sigma_{-i\beta/2}(v_n) \right), \quad \sigma_{-i\beta/2}(v_n) = n^{\beta/2} v_n. \quad (3.11)$$

It identifies a thermal half-step and the fixed factor $n^{-1/2}$. Imaginary-time continuation is not unitary real-time propagation. The physical input/output normalization that would implement (3.11) is still to be derived.

There is a second, independent distinction. Since $\sigma_t(P_n) = P_n$, a native real-time correlation of two such projections is constant:

$$\varphi_\beta(P_m \sigma_t(P_n)) = \varphi_\beta(P_m P_n). \quad (3.12)$$

The number $\log n$ in $\sigma_t(v_n) = n^{it} v_n$ is an energy difference or dilation label. It has not thereby become the causal delay $\log n$ in (2.7). That conversion is a requirement on a coupling to propagating signals, not a consequence of equilibrium positivity.

4 Ordinary local norms and two assembly limitations

4.1 A radial realization of the diagonal state

For one prime ℓ , normalize additive Haar measure dx on $\mathbb{Q}_\ell$ by $\text{vol}(\mathbb{Z}_\ell) = 1$ and define

$$d\mu_{\beta,\ell}(x) = C_{\beta,\ell}|x|_\ell^{\beta-1}dx, \quad C_{\beta,\ell} = \frac{1 - \ell^{-\beta}}{1 - \ell^{-1}}. \quad (4.1)$$

This is the local scaling law of the equilibrium diagonal [2, Section 2]. The shell $v_\ell(x) = m$ has Haar mass $(1 - \ell^{-1})\ell^{-m}$. Consequently its mass under (4.1), for $m \geq 0$, is $(1 - \ell^{-\beta})\ell^{-m\beta}$, exactly (3.9). Thus

$$\mu_{\beta,\ell}(\mathbb{Z}_\ell) = 1, \quad \mu_{\beta,\ell}(aE) = |a|_\ell^\beta \mu_{\beta,\ell}(E), \quad \mu_{\beta,\ell}(\ell^r \mathbb{Z}_\ell) = \ell^{-r\beta} \quad (r \geq 0). \quad (4.2)$$

The unit shell supplies the primitive probability $1 - \ell^{-\beta}$.

The normalized half-density in the fixed Haar Hilbert space is

$$u_{\beta,\ell}(x) = \sqrt{C_{\beta,\ell}}|x|_\ell^{(\beta-1)/2} 1_{\mathbb{Z}_\ell}(x), \quad \|u_{\beta,\ell}\|_2 = 1. \quad (4.3)$$

This realizes the diagonal state in an ordinary positive norm. It is not claimed to intertwine the full Bost–Connes representations at different temperatures.

Weighted dilations on $L^2(\mu_{\beta,\ell})$ and the canonical half-density transport are

$$D_a^{(\beta)} f(x) = |a|_\ell^{\beta/2} f(ax), \quad T_\beta f(x) = \sqrt{C_{\beta,\ell}}|x|_\ell^{(\beta-1)/2} f(x). \quad (4.4)$$

The scaling law makes $D_a^{(\beta)}$ unitary, and cancellation of the weights gives

$$T_\beta D_a^{(\beta)} T_\beta^{-1} = D_a^{(1)}. \quad (4.5)$$

Therefore a pure measure change does not alter the unitary dilation action after this canonical transport. The parameter enters the state and its observables.

Direct shell integration gives an actual native correlation:

$$\langle u_{\beta,\ell}, D_{\ell^r}^{(1)} u_{\beta,\ell} \rangle = \ell^{-\beta|r|/2}, \quad r \in \mathbb{Z}. \quad (4.6)$$

Indeed the common support has valuation $m \geq \max(0, -r)$ and the integrand summed over shells is $(1 - \ell^{-\beta})\ell^{-\beta m - r\beta/2}$. Its geometric sum proves (4.6). This is different from c_{ℓ^r} : primitive projection and centering remain additional readout choices, even though the local state norm is known.

4.2 Failure of the naive common product vacuum

At the reference temperature $\beta = 1$, $u_{1,\ell} = 1_{\mathbb{Z}_\ell}$. The affinity of the two normalized local vectors is

$$h_\ell(\beta) = \langle u_{1,\ell}, u_{\beta,\ell} \rangle = \frac{\sqrt{(1 - \ell^{-1})(1 - \ell^{-\beta})}}{1 - \ell^{-(\beta+1)/2}}. \quad (4.7)$$

In the incomplete tensor product based on the reference vectors, embed a finite set of changed local states as

$$\Omega_{\beta,S} = \bigotimes_{\ell \in S} u_{\beta,\ell} \otimes \bigotimes_{\ell \notin S} u_{1,\ell}. \quad (4.8)$$

Proposition 4.1 (A scoped product-space obstruction). *For each fixed $0 < \beta < 1$, the finite-place vectors (4.8) are not Cauchy as S exhausts the primes. Their shift tangents at $\beta = 2\omega = 1$ also have unbounded norm.*

Proof. For fixed $\beta < 1$, expansion of (4.7) at large ℓ gives

$$1 - h_\ell(\beta) \sim \frac{1}{2}\ell^{-\beta}. \quad (4.9)$$

The prime sum diverges, since $\ell^{-\beta} \geq \ell^{-1}$ and $\sum_\ell \ell^{-1}$ diverges. Hence $\prod_\ell h_\ell(\beta) = 0$. For $S \subset T$ the normalized vectors satisfy

$$\|\Omega_{\beta,T} - \Omega_{\beta,S}\|^2 = 2 \left(1 - \prod_{\ell \in T \setminus S} h_\ell(\beta) \right). \quad (4.10)$$

For every fixed finite S , the right side approaches 2 as T grows, which excludes the Cauchy property.

For the tangent, express the normalized local vector in its shell basis as $a_m(\beta) = \sqrt{(1-q)q^m}$, where $q = \ell^{-\beta}$. Differentiation gives

$$\partial_\beta a_m = \frac{\log \ell}{2} \left(\frac{q}{1-q} - m \right) a_m. \quad (4.11)$$

The geometric variance is $q/(1-q)^2$. Taking account of $\partial_\omega = 2\partial_\beta$ at $\beta = 1$ yields

$$\left\| \partial_\omega \Omega_{2\omega,S} \Big|_{\omega=1/2} \right\|^2 = \sum_{\ell \in S} (\log \ell)^2 \frac{\ell}{(\ell-1)^2}. \quad (4.12)$$

Each local derivative is orthogonal to its normalized real vector, so the product derivative has no cross terms. The last prime sum diverges. $\square$

Some finite-place values illustrate the calculation:

Primes	Largest	Overlap, $\beta = .5$	Overlap, $\beta = .8$	Tangent norm ²
4	7	0.805625	0.978404	3.41186
25	97	0.398762	0.924811	10.9944
168	997	0.0444597	0.819804	23.9650
1,229	9,973	$8.01090 \cdot 10^{-5}$	0.658092	42.3681
9,592	99,991	$1.73891 \cdot 10^{-12}$	0.454913	66.1381

The proof is the divergent-series argument, not the finite table.

This obstruction concerns the stated Haar-based embedding. It does not invalidate the established positive KMS states, whose representations are supplied by their own construction. Nor does it exclude a different common representation, a correlated archimedean coupling, or a different radiation map. Finite-observable thermal expectations can be differentiable even when this vector family is not. No claim about the canonical Sonin embeddings studied elsewhere in the project follows from this calculation.

4.3 The bare Euler delay train is not contractive

A second tempting assembly would turn the numbers $\varphi_\beta(B_{n,\beta})$ directly into delayed scalar amplitudes:

$$R_\omega = I + \sum_{n \geq 2} c_n(\omega) S_{\log n}, \quad (4.13)$$

where $S_v f(u) = f(u - v)$ with zero extension. Each finite observation window sees only finitely many shifts, so the following exclusion requires no definition of the infinite sum as a global bounded operator.

Proposition 4.2 (Bare Euler readout exclusion). *For $\omega > 0$, the compression of (4.13) to any window $\log 2 < L < \log 3$ is not a contraction in ordinary $L^2(0, L)$.*

Proof. Only $I + c_2 S_{\log 2}$ appears on this window. Choose a unit pulse supported on $(0, \varepsilon)$, where $0 < \varepsilon < \min(\log 2, L - \log 2)$. The direct and delayed copies are disjoint and contained in the window. Thus

$$\left\| 1_{(0, L)} R_\omega f \right\|_2^2 = 1 + c_2(\omega)^2 > 1. \quad (4.14)$$

□

The squared norm is approximately 1.06066017177982 at $\omega = 1/4$, and 1.25 at $\omega = 1/2$. Positive contractive observables at individual labels do not control the energy of their coherent sum.

Equation (4.14) does not exclude the completed response $A_\omega Z_\omega$. The same bare Euler failure occurs at the known modular half shift, where the archimedean factor is essential [10]. The next physical test must therefore include that sector and its flux, rather than infer passivity from the signs or individual sizes of the arithmetic weights.

5 The next test: a finite-place physical interface

The coefficient calculation selects a concrete next candidate: the positive sector (3.8) for $S = \{2, 3\}$, coupled to an archimedean radiation channel. It has native observables P_2 , P_3 and P_6 , and its equilibrium state exists at every relevant temperature. The mathematical task is now to specify and derive the coupling, not to invent another list of arithmetic weights.

The first comparison should include $n = 2, 3, 4, 6$. The first two test the primitive amplitudes, $n = 4$ tests repetition, and $n = 6$ tests the combination of distinct primes. A successful model must meet four linked requirements:

1. An explicit Hamiltonian or boundary coupling, together with its state preparation and measured output, must convert the native arithmetic labels into physical propagation delays $\log n$.
2. The same construction must derive the centered amplitudes $n^{(\beta-1)/2} \varphi_\beta(P_n)$ and their parameter derivative, including $c'_2(1/2) = (3/2) \log 2$. A temperature-dependent detector chosen to equal $B_{n,\beta}$ is not by itself that derivation.
3. The archimedean sector must produce A_ω in (2.5) with $\beta = 2\omega$, including the rational completion as well as the gamma factor.
4. The full assembly must have an ordinary radiation-energy balance that accounts for internal storage, later output and any additional channels. It must not define its positive norm by pulling back the desired arithmetic transfer.

Assigning a filter equal to A_ω times a finite Euler product would merely restate the target. A finite-place model need not be lossless if omitted channels carry explicitly accounted flux; an unexplained gain or deficit cannot be treated as the desired scalar response.

Only after this finite interface is derived should one seek an infinite-place common-space limit. Proposition 4.1 identifies a specific embedding that cannot supply it. The full KMS construction

remains available, but an intertwiner from that construction to the required fixed radiation norm is an additional obligation.

The zero-shift endpoint requires a separate check as well. For each fixed $n > 1$, $c_n(\omega) \rightarrow 0$ as $\omega \downarrow 0$, while $c_1 = 1$. This is the correct coefficientwise identity limit. Meanwhile, the geometric thermal distribution at each prime spreads to arbitrarily high occupation as $\beta \downarrow 0$. Coefficientwise convergence is therefore not yet a strong state limit or a strong scattering limit. The current result supplies an independently motivated arithmetic probability law and precisely locates the physical work that remains.

6 Earlier tests: a brief account of the exclusions

The earlier investigations narrowed the required physical readout. Their exclusions concern the specified constructions, rather than entire field theories or all possible couplings. The detailed calculations, sources and diagnostics remain in the cited notes.

WZW evolution and the reflected collar. The deterministic $SU(2)$ level-2 pilot gave an explicit selected block and a positive integrated tensor-norm balance for a constant Loewner driver. General moving drivers did not preserve contraction, and the raw smeared kernel did not approach the required identity operator [5]. A subsequent full-module collar had a positive norm balance and a strong identity limit, but its fixed reflected readout was self-adjoint. A bounded self-adjoint causal scalar operator on an interval is a multiplication operator, which excludes the nontrivial arithmetic Volterra response. The selected primary also had the wrong gamma-tower weights and no first-prime delay [6].

Direct arithmetic Loewner sources. The completed xi logarithmic derivative generated genuine normalized Loewner evolutions after a safe spectral translation, with an exact formula recovering the arithmetic source. The arithmetic information was supplied through xi itself. Removing the safe translation did not follow from the geometric construction or its weighted bound, so this did not furnish an independently motivated physical readout [7].

Brownian bridge and first-passage readouts. Bridge-range moments supplied an independent scalar xi identity. The specified positive first-passage readout was causal and had an ordinary norm balance, but its Laplace response was not the required quotient: its kernel had the wrong front behavior and no arithmetic singularity at $\log 2$. A logarithmic readout recovered the boundary phase without establishing its full-family causality. The distinction was between the native moment identity and the measured response [8, 9].

Modular scattering and a cusp load. Passing to exact one-forms supplied the rational completion of modular scattering and realized $K_{1/2}(p) = \xi(p)/\xi(p+1)$ with the stated ordinary radiation norm and fixed-shift causality. This remains a positive benchmark. The tested real, frequency-independent cusp delta load was lossless and causal but had the wrong arithmetic shift tangent, so it did not extend that benchmark to the required family [10].

Fractional clocks and scalar holonomy. A positive weighted diffusion field varied the leading front exponent, but its scalar clock composition with the modular response failed the subleading term, boundary modulus, fixed prime delays and identity endpoint. An analytical exclusion covered scalar complete-Bernstein clocks, not general coherent couplings [11]. A flat scalar unitary twist

on the original modular surface also supplies no continuous control: $S^2 = R^3 = 1$ allows only six characters, and trivial cusp holonomy leaves only the trivial character. Nontrivial cusp holonomy removes the open constant mode [4].

These tests motivate changing primitive orbit weights and the archimedean response together. They do not establish that a Bost–Connes coupling will succeed; they identify the failures that its actual preparation and readout must avoid.

A Numerical evidence, provenance and acknowledgements

A.1 The current diagnostic

The orbit-weight investigation has an existing record of 87 passing controls: 17 exact integer cases and 70 floating diagnostics. The manuscript integrates those results; it does not claim a new experiment or numerical proof. Its program and record are linked below, relative to the manuscript directory:

- [Orbit-weight diagnostic program](#).
- [87-case numerical record](#).
- [Same-assistant mathematical and numerical audit](#).

The exact controls enumerate primitive residue tuples in dimensions 1, 2 and 3 for moduli 2, 3, 4 and 6, check divisibility-projection products, and enumerate scalar character phases. These finite checks do not replace the general algebraic arguments in the text.

The floating controls compare the Euler products with a Möbius convolution, a full zeta quotient with its truncated Dirichlet series, and a differentiated finite Euler product with its prime-power source. They also check finite-place state masses, local half-density norms, affinities, native dilation correlations, shift tangents, and the piecewise pulse-energy calculation. The affinity table in Section 4 retains only compact samples, not a large generated dataset.

The recorded environment is Python 3.10.0 with mpmath 1.3.0 at 50 decimal digits. Shell sums run through occupation 360, the Dirichlet comparison through $n = 1500$, and the prime table through 100,000. For the comparison at the positive real Laplace parameter $p = 8$, the elementary bound $0 < c_n(\omega) \leq n^{\omega-1/2}$ gives, with $\alpha = p + 1/2 - \omega > 1$,

$$0 \leq Z_\omega(p) - \sum_{n \leq N} c_n(\omega) n^{-p} \leq \frac{N^{1-\alpha}}{\alpha - 1}. \quad (\text{A.1})$$

Evaluating this proven bound in floating arithmetic does not make the comparison an interval certificate. No xi-zero table or external dataset is used. The record stores a SHA-256 program hash, which establishes file identity rather than correctness.

Earlier suites are separate: the WZW pilot has 72 controls, the direct arithmetic source 60, the collar 60, Brownian readout 51, modular scattering 65, and the fractional cusp test 80. Their notes document their exact scopes. The two earlier general-shift quadrature implementations were corrected and fully replayed during the modular test [10, Section 8]; the analytical exclusions were unchanged. Those historical suites have not been rerun merely for this editorial revision.

A.2 Research and document provenance

The principal source of the current argument is [4]. The earlier notes are cited in Section 6 so that the retrospective need not repeat their derivations. The local README and build record identify the active TeX inputs, delivered PDF, source notes, numerical records and review. The manuscript compiles from local TeX inputs; it does not import the parent Wilson–Loewner manuscript. The milestone index records this live revision without creating a dated manuscript snapshot.

The coefficient proofs and scoped exclusions are analytical arguments. The numerical controls do not prove the existence of a causal physical interface, the full-family unweighted contraction, or RH. The next construction is the finite-place interface in Section 5, rather than a numerical search for zeros.

A.3 Acknowledgements and review status

This manuscript was drafted for Edward Baker with substantial LLM assistance from GPT-6 (Codex, according to the developer-provided model identity), including mathematical synthesis, algebraic calculations, diagnostic programming, source checking and document preparation. The reasoning-effort setting was not exposed and is not inferred. The research and manuscript audits were performed by the same assistant; no independent specialist review is claimed. Established results are attributed to their sources, and the unresolved centering, causality, completion and norm requirements are stated explicitly.

References

- [1] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, *Selecta Math. (N.S.)* **1** (1995), 411–457. The equilibrium facts used here are also stated in [2].
- [2] S. Neshveyev, *Von Neumann algebras arising from Bost–Connes type systems* (2009), [arXiv:0907.1456](#), Introduction and Section 2.
- [3] M. Suzuki, *A canonical system of differential equations arising from the Riemann zeta-function*, [arXiv:1204.1827](#), version 2.
- [4] Project research note, *Changing arithmetic orbit weights: a Bost–Connes thermal test*, 23 September 2026. [Detailed derivations, diagnostic and audit links](#).
- [5] Project research note, *$SU(2)$ level 2: a completed deterministic boundary Loewner pilot*, 22 September 2026. [Detailed WZW calculation and controls](#).
- [6] Project research note, *Bounded arithmetic readout test: WZW collar, spectral weights and first delay*, 23 September 2026. [Full-module readout exclusion and primary weights](#).
- [7] Project research note, *The arithmetic source and an actual generalized Loewner evolution*, 22 September 2026. [Source, norm and continuation analysis](#).
- [8] Project research note, *Independent arithmetic-source search*, 23 September 2026. [Primary-literature comparison and candidate selection](#).
- [9] Project research note, *Brownian bridge readout test: first passage, logarithmic range and phase*, 23 September 2026. [Native response, norm balance and arithmetic exclusion](#).
- [10] Project research note, *Modular Hodge scattering and a bounded cusp-coupling test*, 23 September 2026. [Half-shift benchmark, cusp-load exclusion and numerical correction](#).
- [11] Project research note, *Continuous exponent and fractional cusp-field test*, 23 September 2026. [Positive memory field and scalar-clock obstruction](#).